

11 D Resultant force towards the sun equals to ma_c .

12 D GPE of orbiting satellite $E_p = -G \frac{Mm}{r}$

Gravitational force = centripetal force

$$\Rightarrow G \frac{Mm}{r^2} = m \frac{v^2}{r}$$

$$\Rightarrow E_k = \frac{1}{2}mv^2 = \frac{1}{2}G \frac{Mm}{r}$$

$$\Rightarrow E_p = -2E_k$$

$$E'_p = E_p + E$$

$$\Rightarrow -2E'_k = -2E_k + E$$

$$\Rightarrow E'_k = E_k - \frac{E}{2}$$

13 A Total number of moles of gas is constant. Since $PV = nRT$, we take the initial number of